

Relational Admissibility as a Computational Primitive

Oblivious Compute distributes a single admissibility function across a set of independently state-maintaining observers within a state space Ω . No observer requires a computationally designated neighbor; each evaluates presented state from its own position. Together, these local determinations form a matrix of relations across the observer set.

$$A : \Omega \times \Omega \rightarrow \{0, 1\} \quad A(s, x) \in \{0, 1\}$$

The same presented state may therefore be admissible from one observer position and inadmissible from another.

$$s_i \neq s_j \quad \Rightarrow \quad A(s_i, x) = 1, \quad A(s_j, x) = 0$$

Across the observer set, let Σ denote relational symmetry within the resulting matrix of relations. That symmetry constitutes the computational field $\mathcal{F}$.

$$\mathcal{F} \equiv \Sigma(s_1, s_2, \dots, s_n)$$

No observer contains the field. The field contains no state. It exists only through the symmetry between independently maintained states.

What's the Difference

By using relational symmetry, Oblivious Compute moves the computational locus out of the observer and into a stateless field, a construction that may constitute a new computational primitive.

Testbeds

Service discovery · adversarial networks · multiplayer state · distributed monetary systems · distributed AI clusters · delay-tolerant and interplanetary systems · modular quantum systems